I. Introduction

In recent years, global disparities in adult mortality rates have drawn increasing attention in public health and policymaking. Adult mortality, which for our purposes refers to the number of deaths among individuals aged 15 to 60, is a key indicator of the health and social development of a nation. Understanding what drives these differences across countries is crucial for designing effective interventions by targeting known factors which have a greater effect on adult mortality rates. This project aims to investigate how various socioeconomic, health, and immunization-related factors influence adult mortality across 100+ countries between 2000 and 2015.

Research Question: To what extent can the adult mortality rate of a nation be correlated to the socio-economic, biological, and environmental factors at play such as development status, alcohol consumption, healthcare expenditure, education level, and others.

The response variable for this study is adult mortality rate, a continuous variable representing deaths per 1,000 adults aged 15 to 60.

The predictor variables include year, development status, life expectancy, alcohol consumption, percentage of GDP spent on healthcare, vaccination coverage (Hepatitis B, Polio, and Diphtheria), measles rates, HIV/AIDS deaths, BMI, total healthcare expenditure, GDP, population, and average years of schooling.

Linear regression is well-suited for this investigation for 3 main reasons. Firstly, a linear regression model allows us to quantify the direct relationship between adult mortality and multiple predictor variables using clear statistical outputs like coefficients and confidence intervals. Secondly, focusing on interpretability, we can aim to describe which variables contribute most to adult mortality and by how much, rather than constructing a purely predictive model. Finally, the assumptions of linear regression are generally appropriate for this dataset, with a continuous response variable and mostly continuous or binary predictors. Where needed, we will diagnose and address violations (e.g., non-linearity, heteroscedasticity, etc.) using residual plots and model summaries and later, improving our model using various transformations.

In short, this project provides not just an academic exercise in regression modeling, but also practical insight into which factors most strongly drive adult mortality to inform global health strategies, resource allocation, and country-specific interventions aimed at improving the well-being of working-age populations.

II. Data Description

Data Source:

The data, "Life Expectancy (WHO)," was sourced from Kaggle to compare various factors from different nations and their effect on life expectancy. This data was collected by the World Health Organization from national surveys, censuses, and national records from various nations

between 2000-2015. This data was intended to be used for life expectancy research thus, we are using the data with a different purpose in mind.

Response Variable:
We chose adult mortality rates, which represent the number of deaths per 1,000 individuals aged 15 to 60, as our response variable. This choice is suitable for linear regression analysis as it is continuous and easily observable. Further, our dataset contains no extreme outliers that would skew out results. Below is a statistical summary of the variable. We can assume that the distribution is right-skewed due to the mean being greater than the median.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.0	77.0	148.0	168.2	227.0	723.0

Predictor Variables:

```
   country               year         status          life_expectancy adult_mortality
 Length:1649        Min.   :2000   Length:1649        Min.   :44.0    Min.   :  1.0
 Class :character   1st Qu.:2005   Class :character   1st Qu.:64.4    1st Qu.: 77.0
 Mode  :character   Median :2008   Mode  :character   Median :71.7    Median :148.0
                    Mean   :2008                      Mean   :69.3    Mean   :168.2
                    3rd Qu.:2011                      3rd Qu.:75.0    3rd Qu.:227.0
                    Max.   :2015                      Max.   :89.0    Max.   :723.0
    alcohol       percent_expenditure  hepatitis_b       measles            bmi
 Min.   : 0.010   Min.   :    0.00    Min.   : 2.00   Min.   :     0   Min.   : 2.00
 1st Qu.: 0.810   1st Qu.:   37.44    1st Qu.:74.00   1st Qu.:     0   1st Qu.:19.50
 Median : 3.790   Median :  145.10    Median :89.00   Median :    15   Median :43.70
 Mean   : 4.533   Mean   :  698.97    Mean   :79.22   Mean   :  2224   Mean   :38.13
 3rd Qu.: 7.340   3rd Qu.:  509.39    3rd Qu.:96.00   3rd Qu.:   373   3rd Qu.:55.80
 Max.   :17.870   Max.   :18961.35    Max.   :99.00   Max.   :131441   Max.   :77.10
     polio          diphtheria       HIV/AIDS           gdp               population
 Min.   : 3.00   Min.   : 2.00   Min.   : 0.100   Min.   :     1.68   Min.   :3.400e+01
 1st Qu.:81.00   1st Qu.:82.00   1st Qu.: 0.100   1st Qu.:   462.15   1st Qu.:1.919e+05
 Median :93.00   Median :92.00   Median : 0.100   Median :  1592.57   Median :1.420e+06
 Mean   :83.56   Mean   :84.16   Mean   : 1.984   Mean   :  5566.03   Mean   :1.465e+07
 3rd Qu.:97.00   3rd Qu.:97.00   3rd Qu.: 0.700   3rd Qu.:  4718.51   3rd Qu.:7.659e+06
 Max.   :99.00   Max.   :99.00   Max.   :50.600   Max.   :119172.74   Max.   :1.294e+09
   schooling      continent
 Min.   : 4.20   Length:1649
 1st Qu.:10.30   Class :character
 Median :12.30   Mode  :character
 Mean   :12.12
 3rd Qu.:14.00
 Max.   :20.70
```

The preceding graphic contains all the numerical data regarding all our predictors. For this proposal, we focus on exploring the five chosen predictors for preliminary analysis.

The preliminary model includes five predictors: developmental status, alcohol consumption, percentage expenditure, Hepatitis B vaccination, and schooling.

Developmental status is a categorical variable that categorizes a country as developing or developed.

Alcohol consumption is right-skewed, as only a small percentage indicated a higher intake than the others.

Percentage expenditure is also right-skewed and has higher variability, which shows that some countries invest more than others.

A few observations have very low vaccination rates, which pulls the hepatitis B vaccination data lower, though most countries have coverage around 80%.

The distribution of schooling is symmetrical with a mean of 11.9 years.

These variables were selected because they represent important factors which affect adult mortality rates. Also, there are not many extreme outliers, and they are generally well distributed.

III. Preliminary Model Results

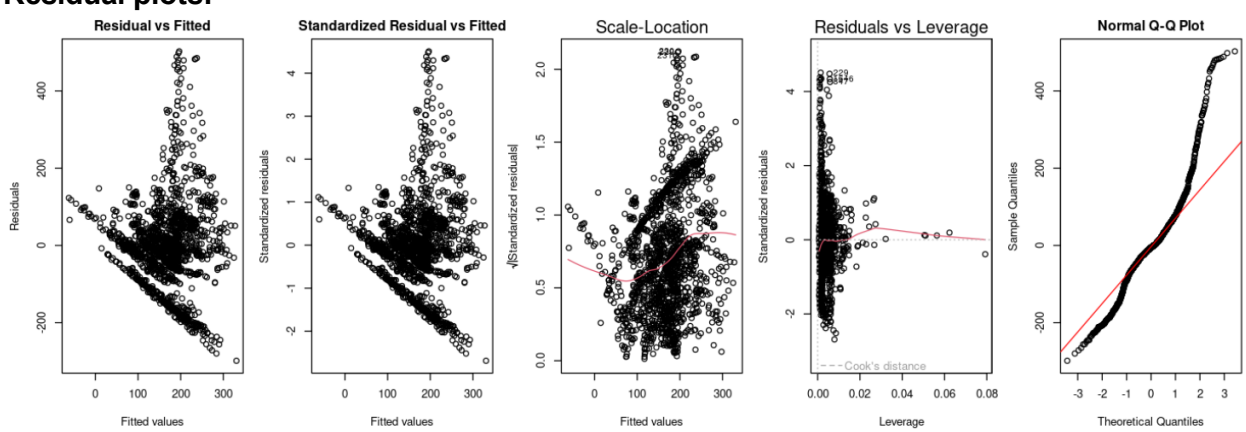
Proposed model:

$$\begin{aligned} &Adult\ Mortality_i \\ &= \beta_0 + \beta_1 I(Status\ is\ Developing)_i + \beta_2 Alcohol_i + \beta_3 Percent\ Expenditure_i \\ &\quad + \beta_4 HepatitisB_i + \beta_5 Schooling_i \end{aligned}$$

Model estimates:

Coefficient	β_0	β_1	β_2	β_3	β_4	β_5
Estimate	340.36	39.97	5.94	-0.0059	-0.216	-17.99

Residual plots:



*These are selected for their ability to illuminate all assumptions of a linear regression model

Residual and Standardized Residual vs. Fitted Values Plots:

The first plot shows that the residuals are not randomly scattered around zero and do not seem to average to zero. The residuals exhibit an observable pattern as a function of the fitted values, specifically an upward u-shaped curve, hence residuals and fitted values are not uncorrelated. Therefore, it is sufficient to conclude that the model violates the linearity assumption. From the second plot, the spread of the residuals grows larger as the predicted value increases, which suggests non-constant variance of the error term across the observations. Therefore, there is

significant evidence to conclude that the model is also violating the homoscedasticity assumption. Similar ideas are reflected in the scale-location plot and the residuals vs. leverage plot.

Q-Q plot:
The Q-Q residual plot shows that the majority of points align with the superimposed line, which suggests that the residuals mostly resemble a normal distribution. However, the right tail of the plot curves sharply upward, causing approximately a quarter of the points to deviate significantly from the line. Because of this skewness, the preliminary model is likely violating the normal errors assumption.

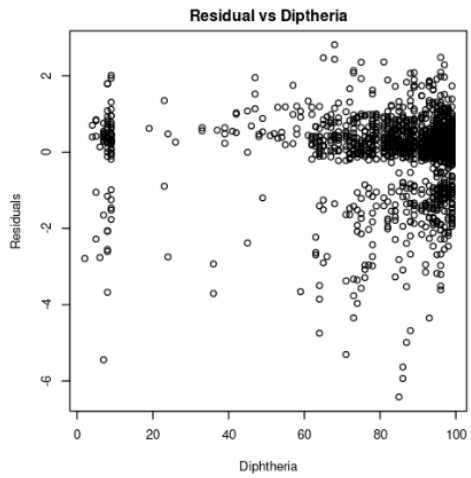
Preliminary Findings:
This preliminary model has identified some key predictors of adult mortality. On average, developing countries have 39.97 more adult deaths per 1000 people than developed countries, a categorical predictor highlighting a disparity in health outcomes between countries with different development statuses. The model also reveals that with other variables held constant, an additional year of schooling, a continuous predictor decreases adult deaths by 17.99 (per 1000 people), confirming that educational outcome is tightly linked to adult mortality. The model also suggests that adult mortality is influenced by many other factors, as greater healthcare expenditure, vaccination coverage, and lower alcohol consumption are all associated with fewer deaths. Overall, there have been good relationships established between our response and predictors.

IV. Model selection

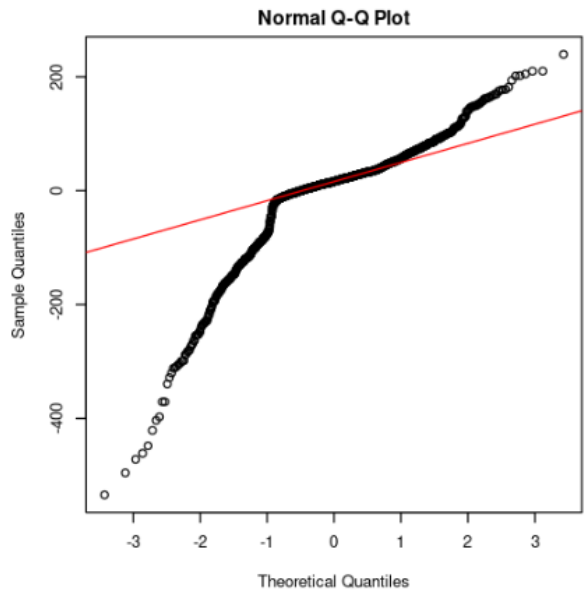
Initial proposed model:

Our initial attempt at a model was simply to use a multiple linear regression model where we ran into initial problems regarding our linear regression assumptions. From here, we went through various changes in our model and data set to contribute to a better performing model.

Transformation Selection
Our primary transformation was to use the Box-Cox transformation to improve the variance and normality of our residuals to better follow linear regression assumptions.



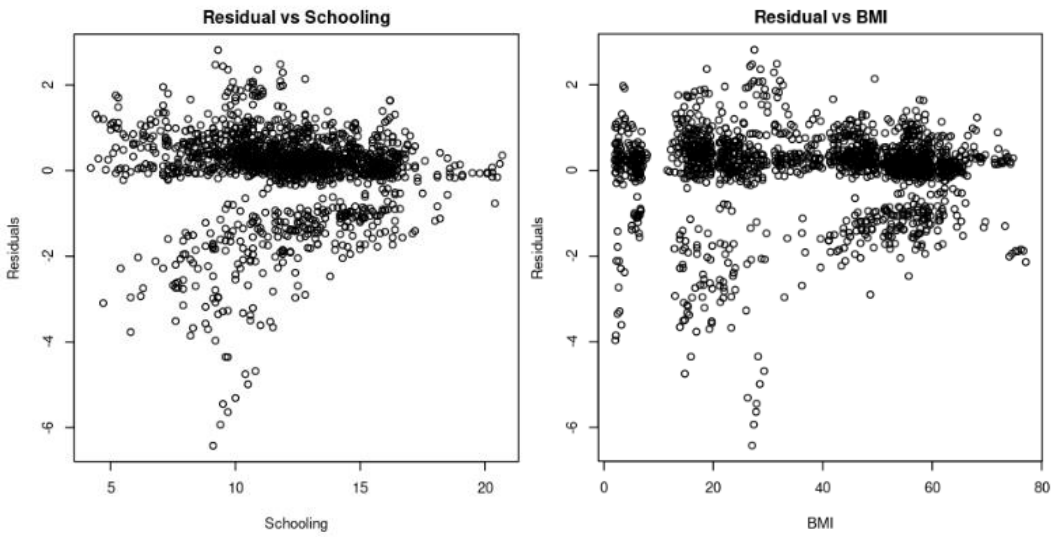
For example, taking one of our initial residuals plots mapping the predictor percentage of Diphtheria vaccination to the standardized residuals, we can see that our residual variance is not consistent through the different values of our predictor and that we have many residual values, particularly some of those of residuals which extend beyond negative six, that would not be indicative of a normal distribution as the spread of their values is far from zero than what we would have expected.



This was our Q-Q plot for our initial full model mapping the standardized residuals to their theoretical quantiles. Clearly, we have a heavy left-skew and a very high right tail, both of which suggest that our residuals are not normally distributed like what we would expect of them.

Outlier Removal:

As we worked further with our model and data, we noticed that no matter what transformations we applied on the different predictor or response variables, there was a segregation in the residual plot for every predictor in that there was a clear-cut border of sorts which separated the general results of the residuals and ones which were vastly different from those. For example, the following residual plots:

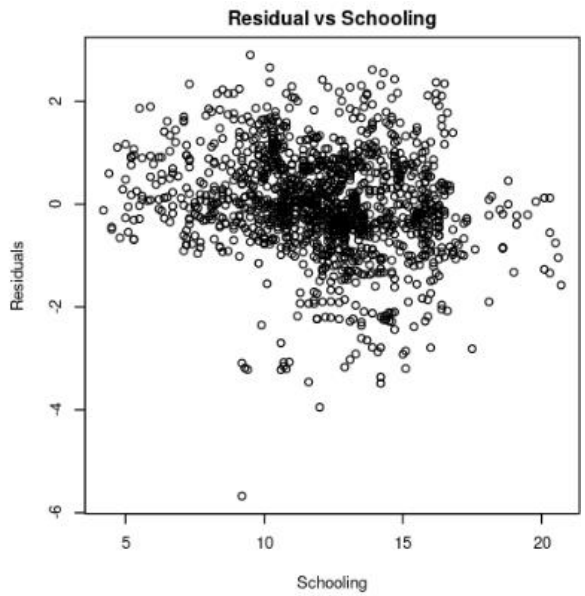


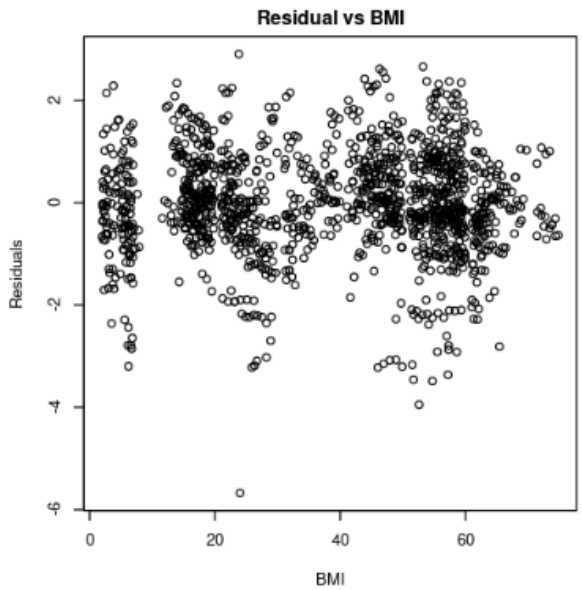
Of schooling and BMI showcases this fact clearly. You can clearly imagine an invisible line below the value zero which separates the bulk of the points to a section of a smaller island of points. Also, this separate island of points seems to be distributed in its own normal distribution with a mean below zero. This would imply that there are a series of data points that are consistently over-predicted, in terms of adult mortality rates, by the model than what the observable data would suggest.

We were able to identify the bulk of these points as belonging to the countries: Lesotho, Eswatini, Zimbabwe, Central African Republic, Chad, Sierra Leone, Mozambique, Guinea-Bissau, Burundi, and Angola. Our model would suggest that these countries tend to exhibit different behaviours than the others that were included in the dataset, namely the majority of countries in the world.

Coincidentally, or perhaps by factors not represented in the dataset, these countries all reside on the continent of Africa, more specifically the regions from the central to the southern parts and the west coast of Africa. Purely based on speculation, and not at all represented in our dataset, this could be due to the lack of institutional depth to properly report adult mortalities - perhaps an attempt to downplay the officially reported adult mortality rates, or simply due to the unique circumstances including socio-economic, geographic, and biological, that these countries possess.

Disregarding these speculations, with these data points removed, we were able to obtain the following residual plots:





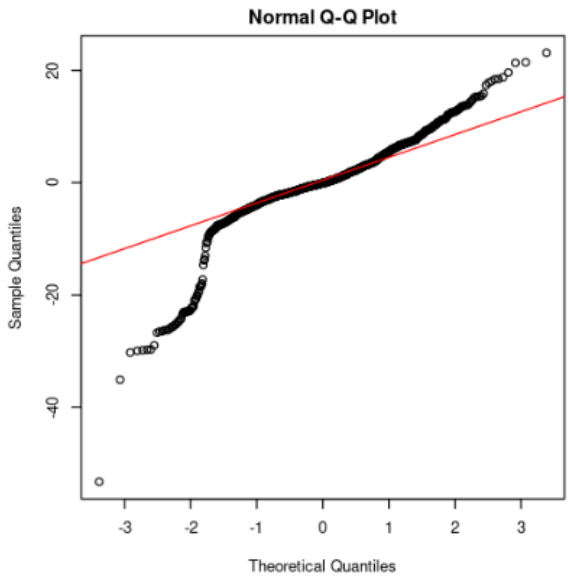
Which shows that many of the outliers have been removed, and that the residuals now are more in line with the regression assumptions.

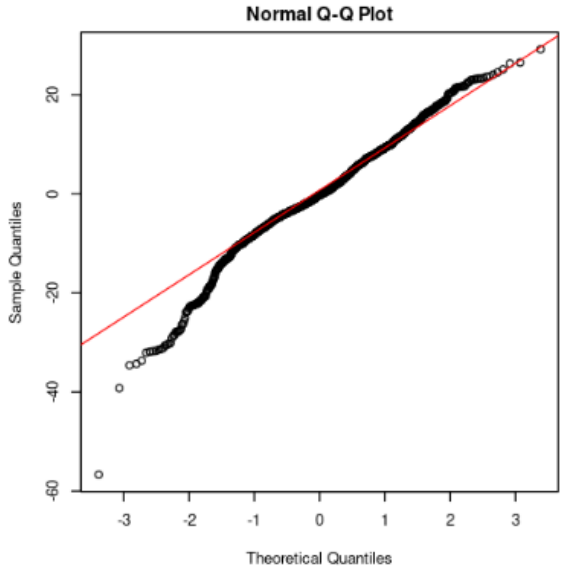
Predictor Inclusions/Exclusions:

Exclusions:
As we worked with our initial full model, we eventually decided to exclude the predictors' life expectancy and year. Also, though not entirely excluded, we decided to condense each individual country into their respective continental regions to simplify the regression model.

The removal of the year predictor was for two reasons. Firstly, the p-values for all levels of the year variable were statistically insignificant, ranging from values of 0.11 to 0.98. Secondly, the different intercepts associated with each year were arbitrary where the year would randomly contribute or decrease the predicted adult mortality rates.

The removal of life expectancy also was incurred for the following two reasons. Firstly, there was a high VIF of over five when it was calculated. Also, when we included life expectancy in our model, we noticed a heavy left skew and a higher right resulting in the following Q-Q plot:





The right one being the model with the predictor removed.

Finally, we removed GDP from the model due to its high VIF, which we assumed to be its collinearity with the percent expenditure predictor. This resulted in an acceptable VIF lower than two for percent expenditure and resulted also in an increased R-squared and adjusted R-squared values.

Inclusions:
We decided to add an interaction term between region and HIV. In the real world, HIV/AIDS is a disease which can be manageable if in a country of a more developed status with greater healthcare infrastructure. However, the same cannot be said about less fortunate countries where HIV/AIDS can be detrimental. As such, despite having the same rates of HIV/AIDS, it is probable that its effect in different countries is drastically different in terms of its effect on adult mortality rates in that country.

V. Final model inference and results

Model Summary				
Term	**Estimate**	**Standard Error**	**P-value**	**95% Confidence Interval**
Intercept	136.9347	8.8669	0	(119.5407, 154.3287)

regionEurope & Central Asia	-22.6099	5.6705	0.0001	(-33.7337, -11.4862)
regionLatin America & Caribbean	-13.1346	5.3829	0.0148	(-23.6942, -2.5749)
regionMiddle East & North Africa	-47.4519	5.9533	0	(-59.1304, -35.7734)
regionNorth America	-49.2989	14.1256	0.0005	(-77.0087, -21.5891)
regionSouth Asia	-5.5915	8.5651	0.5140	(-22.3935, 11.2106)
regionSub-Saharan Africa	66.8517	5.0173	0	(57.0093, 76.694)
statusDeveloping	30.5148	4.6110	0	(21.4695, 39.5601)
alcohol	1.0680	0.4271	0.0125	(0.2301, 1.9059)
hepatitis_b	0.0963	0.0602	0.1098	(-0.0218, 0.2144)
measles	-0.0002	0.0001	0.1298	(-5e-04, 1e-04)
bmi	-0.2384	0.0782	0.0023	(-0.3917, -0.0851)
polio	-0.1787	0.0653	0.0063	(-0.3068, -0.0506)
diphtheria	-0.2512	0.0767	0.0011	(-0.4017, -0.1007)

HIV/AIDS	71.4053	10.0184	0	(51.7524, 91.0583)
GDP	-0.0009	0.0001	0	(-0.0011, -7e-04)
population	0	0	0.2702	(0, 0)
regionEurope & Central Asia:hiv	85.0022	21.3284	0.0001	(43.1627, 126.8417)
regionLatin America & Caribbean:hiv	-48.4951	11.2133	0	(-70.492, -26.4983)
regionMiddle East & North Africa:hiv	-24.7977	11.4898	0.0311	(-47.3371, -2.2584)
regionSouth Asia:hiv	47.8160	39.7302	0.2290	(-30.122, 125.7541)
regionSub-Saharan Africa:hiv	-63.3036	10.0222	0	(-82.9639, -43.6432)

Model results:

This model aims to explore how adult mortality worldwide is influenced by a range of socio-economic, biological, and environmental factors, using a Box Cox transformation on adult mortality. While this transformation substantially improves model validity, it also reduces interpretability as coefficient estimates now represent the effect of each predictor on adult mortality on the transformed scale rather than the original unit. Nonetheless, the direction and significance of the effects can still provide meaningful answers to our research question.

The intercept estimate of the model is 137, which represents the expected value of transformed adult mortality in a developed East Asia & Pacific country, assuming all other predictors have a value of zero. However, this value has limited real-world interpretability, since it is not possible for any country to have a zero population or zero average BMI.

The region variable uses East Asia & Pacific as the reference category. The coefficient estimates for other regions reflect how transformed adult mortality changes in comparison to the baseline, assuming all other predictors are fixed. In addition, since HIV prevalence varies greatly across regions, an interaction term between region and HIV was included in the model. This interaction term captures how the effect of HIV on adult mortality differs across regions. For Latin America & Caribbean, both the estimates of the region and interaction terms are negative, so with all else equal, a country in this region would have a lower transformed adult mortality on average compared to the baseline category regardless of the level of HIV prevalence. In contrast, consider Europe & Central Asia, which has a negative region estimate of –23 and a positive interaction term estimate of 85. This means the region is associated with lower adult mortality at very low HIV levels, specifically less than 0.27. At higher levels, the effect of HIV on mortality increases more steeply in this region than in the baseline.

The status variable categorizes countries as developed or developing, with “developed” being the baseline. The coefficient estimate is 30, meaning with all else equal, on average, a developing country has a higher transformed adult mortality rate than a developed country. This supports the research question by showing that socio-economic classification plays a significant role in determining adult mortality, likely reflecting differences in access to healthcare and quality of life.

Alcohol consumption has a positive coefficient estimate of 1. This means that for every additional liter of alcohol consumed per capita, the transformed adult mortality rate increases by 1 unit. This reinforces the health risks associated with alcohol as well as the increased risks of accidents. On the other hand, BMI has a coefficient estimate of −0.2384, suggesting that a higher average BMI is associated with lower transformed adult mortality.

HIV prevalence stands out as one of the most influential predictors in the model, with an estimate of 71.4. This means that an increase of 1 HIV death per 1000 births causes average adult mortality to increase by 71 units on the transformed scale, holding other factors constant. This directly supports the research question by identifying a biological factor that is strongly correlated to adult mortality.

Other predictors used in the model, including GDP and several public health determinants, have more modest effects. GDP has a near-zero estimate of -0.0009 and its p-value indicates that the predictor is statistically insignificant at the 5% level. This suggests that being a wealthier country alone does not guarantee better health outcomes. Similarly, hepatitis B immunization coverage and number of measles cases both have near-zero coefficients and insignificant p-values, indicating that these variables do not strongly predict adult mortality in this model. Polio and diphtheria immunization rates, however, show small but statistically significant effects, suggesting that higher vaccination coverage is associated with slightly lower adult mortality, supporting the idea that public health infrastructure contributes to better outcomes.

Model performance:

To assess the performance of the model, we examine several key metrics: R-squared, adjusted R-squared, RSE, AIC, BIC, and p-value of the F test.

The multiple R-squared value is 0.7835, meaning that 78.35% of the variation in adult mortality rates is explained by the predictors used in the model. This suggests that the model does a good job capturing the overall patterns in the data. However, since approximately 20% of the variability in adult mortality remains unexplained by the predictors used, the model can potentially be improved by including other relevant predictors.

Since many predictors were used and R-squared can be unreliable when working with high-dimensional data, we also consider the adjusted R-squared. The adjusted R-squared value of 0.7802 is only slightly lower than the unadjusted R-squared, suggesting that there are no redundant predictors and that the model is not excessively complex relative to its explanatory power.

The Residual Standard Error is 42.14, which is the average distance between the observed and predicted values of adult mortality rates over all observations. Given that adult mortality rates in the dataset span a wide range, an average deviation of 42.14 from the regression line appears reasonable.

The AIC and BIC values of the final model are 14523.15 and 146543.84, respectively. In comparison, the initial model fitted without transforming the response and before removing some predictors yielded AIC and BIC values of 14066.69 and 14255.61. Although the lower AIC and BIC suggest that the initial model is preferable, it suffers a key drawback which is violation of the normality assumption. The final model, on the other hand, is a lot simpler and ensures all regression assumptions are satisfied. Considering that the difference in AIC and BIC values between the two models are relatively small compared to the improvement in model validity, the final model is still preferred.

Finally, the F-statistic is 238.4 with a p-value less than 2.2e-16, meaning that the null hypothesis is rejected at the 5% significance level, and the regression model is preferred over the null model. In other words, at least one of the predictors contributes meaningfully to explaining adult mortality.

VI. Discussion and conclusion

After performing a Box Cox transformation on the response variable and removing outliers as necessary, the final model satisfies all regression assumptions including linearity, homoscedasticity and normality of errors. An analysis of several key metrics including R-squared, adjusted R-squared, RSE, AIC, BIC and the F test suggests that the final model is a

strong fit and has improved significantly from previous models. In summary, results reveal that the most powerful predictors of adult mortality in the model are development status, HIV prevalence, alcohol consumption, and regional context. In addition to the individual effect of each predictor, the interaction between different predictors, particularly region and HIV prevalence, further emphasizes the importance of contextual relationships in explaining the variability in the response. These findings strongly support the research question by showing that adult mortality is not random but rather systematically shaped by a combination of socioeconomic, biological, and environmental factors.

One of the key takeaways from this analysis is the significant effect of a country's development status on its adult mortality rate. The model shows that, all else equal, developing countries consistently exhibit higher adult mortality rates than those that are developed. This result reflects deep-rooted disparities and emphasizes the importance of a well-rounded development in not just economic performance but also in healthcare, infrastructure, and social protections to reduce mortality.

Another key finding is that HIV prevalence stood out as the most powerful predictor even after accounting for differences in region, economic status, and other public health determinants. This aligns with the common understanding that HIV is a major contributor to adult mortality, especially in countries where prevention and treatment programs are not well established. Furthermore, the model also tells us how this effect varies across different regions. By including an interaction term between region and HIV, model results reveal that contrary to common presumptions, less economically developed countries do not necessarily suffer more than wealthier countries from increases in HIV prevalence. For example, in Sub-Saharan Africa, the increase in adult mortality associated with HIV is smaller than in East Asia & Pacific. This might be due to the fact that Sub-Saharan Africa has a long history of being impacted by HIV and hence has a lot more targeted HIV programs in place. On the other hand, in Europe & Central Asia, the model shows that adult mortality rises much more steeply with higher HIV rates. This suggests that in some regions, though more economically developed, health systems may not be as well-equipped to handle increases in HIV prevalence.

The study has provided strong suggestions for lowering adult mortality rates around the world. First, countries with higher average years of schooling had lower adult mortality rates, demonstrating the importance of education. This implies that by expanding people's access to information and resources, investing in a nation's educational system can raise life expectancy and general health. In developing nations, this link is evident. HIV was a significant cause of adult mortality as well, though its impact differed by location. Better access to treatment and more robust health systems can help lower mortality in regions like Sub-Saharan Africa, where HIV prevalence is high. Early testing and prevention should be the primary priorities in other areas where HIV is less prevalent but has a greater impact. Another recommendation is controlling alcohol intake, as nations of all economic backgrounds are impacted by alcohol use, which has also been connected to increased adult mortality. Public health initiatives should therefore concentrate on decreasing harmful drinking and increasing knowledge of its consequences. These suggestions, which are grounded in the study's findings, demonstrate

that social systems and health behavior both have significant effects on adult mortality and should be taken into account when planning public health initiatives.

Future research could enhance this analysis in a number of ways. To satisfy model assumptions, the response variable was changed using the Box-Cox method, which complicated the interpretation of the findings. Future studies could use models appropriate for various data types or attempt more straightforward transformations, such as employing logs. Another improvement that could be made is adding other factors that impact mortality, like political conflict, urbanization, gender inequality, and healthcare quality. We could also consider employing techniques that minimize the impact of outliers without removing them. Lastly, more flexible models or those that take into account regional and national variations may increase accuracy and capture more intricate relationships.

VII. Bibliography

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Global health observatory. (n.d.). Retrieved from https://www.who.int/data/gho